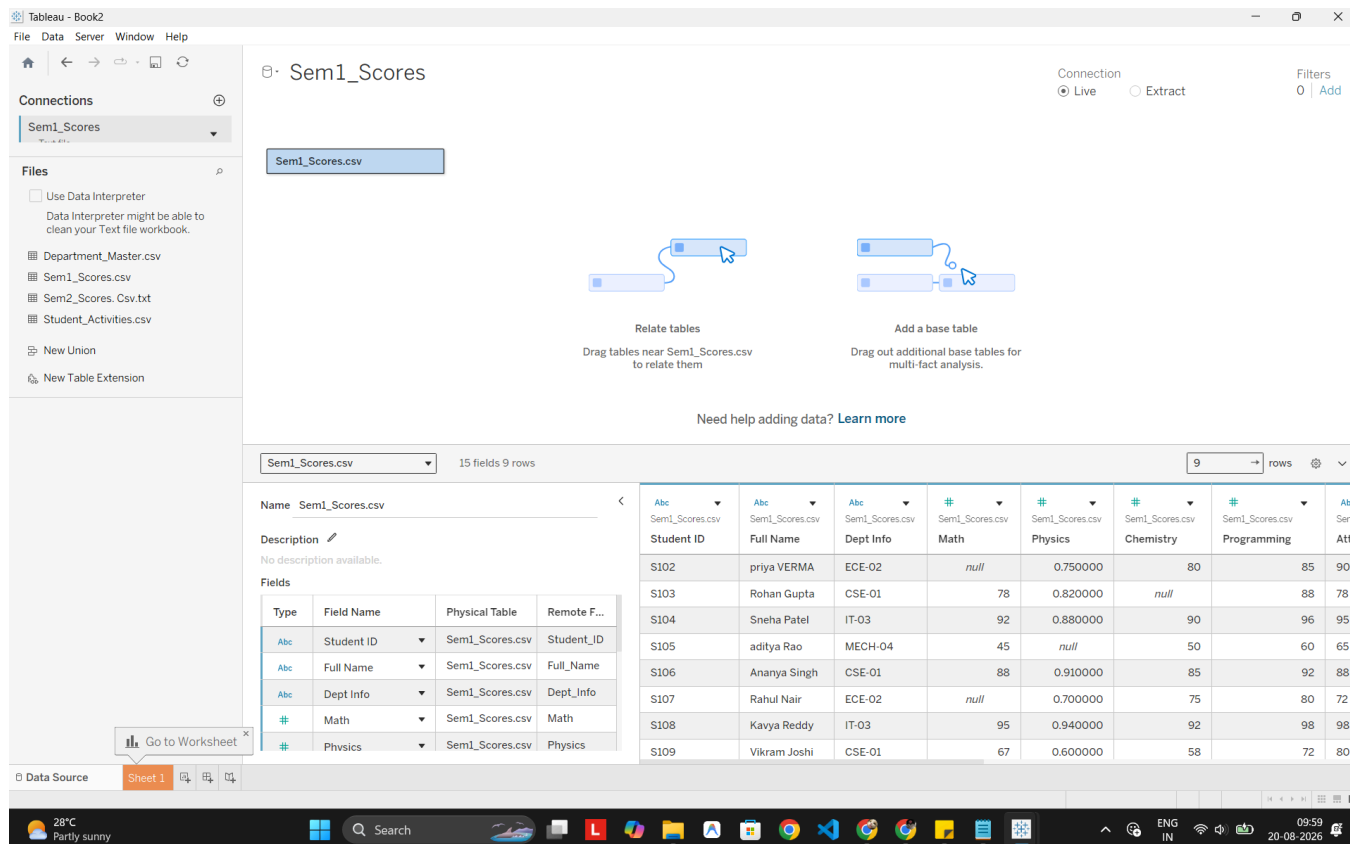


- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

Dataset: Custom Designed Students Dataset [csv files]

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table:
 - o Observe missing entries (Null) in Math, Physics, and Chemistry.



- o Note that Physics and Attendance are classified as text (Abc) due to the trailing % characters.

Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

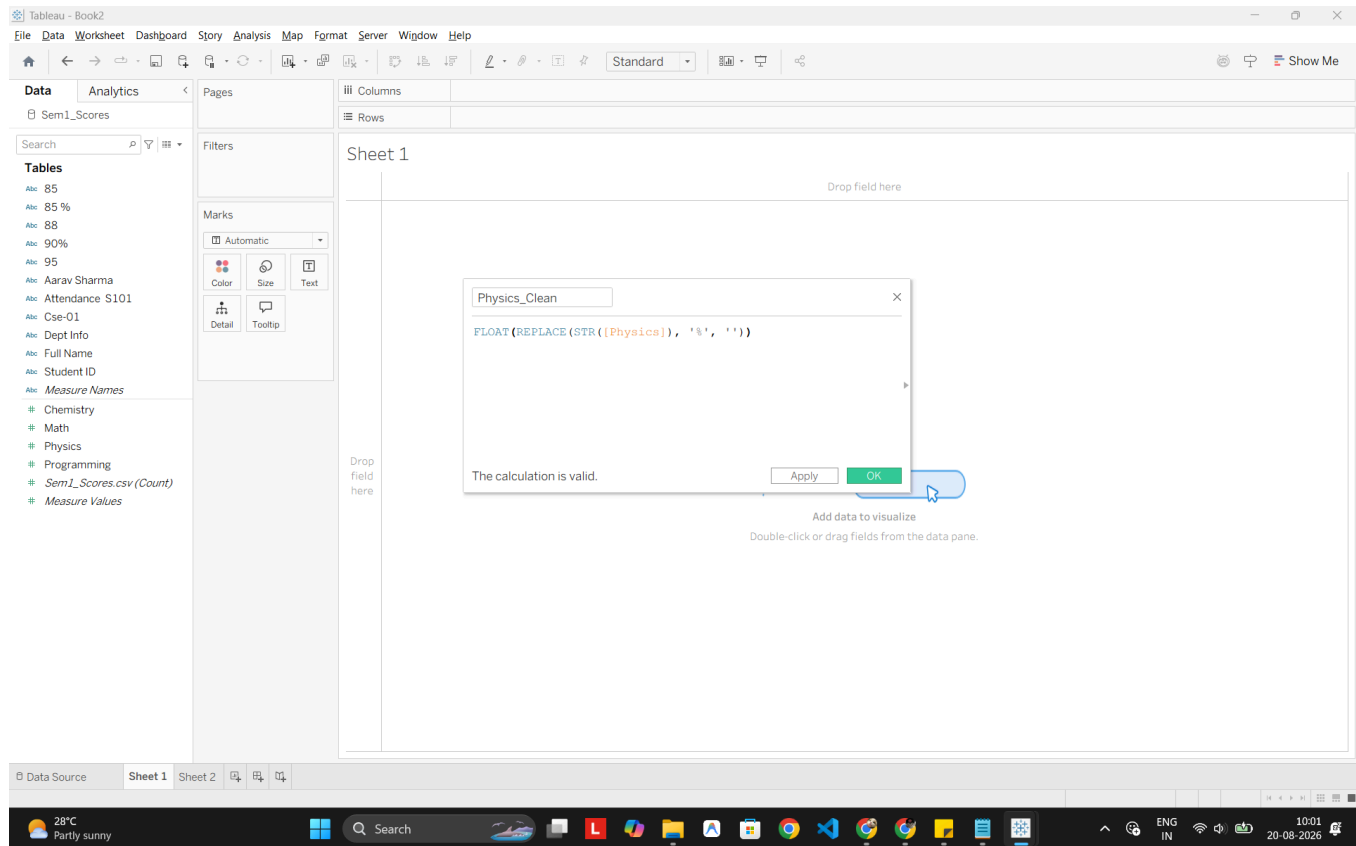
1. Navigate to Sheet 1 (bottom-left tab).
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....
3. Create the cleaned measure for Physics:
 - o Field Name: Physics_Clean
 - o Tableau Formula:

Code snippet

`FLOAT(REPLACE(STR([Physics]), '%', ''))` o Click OK.

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Subject: Data Visualization with Power BI and Tableau



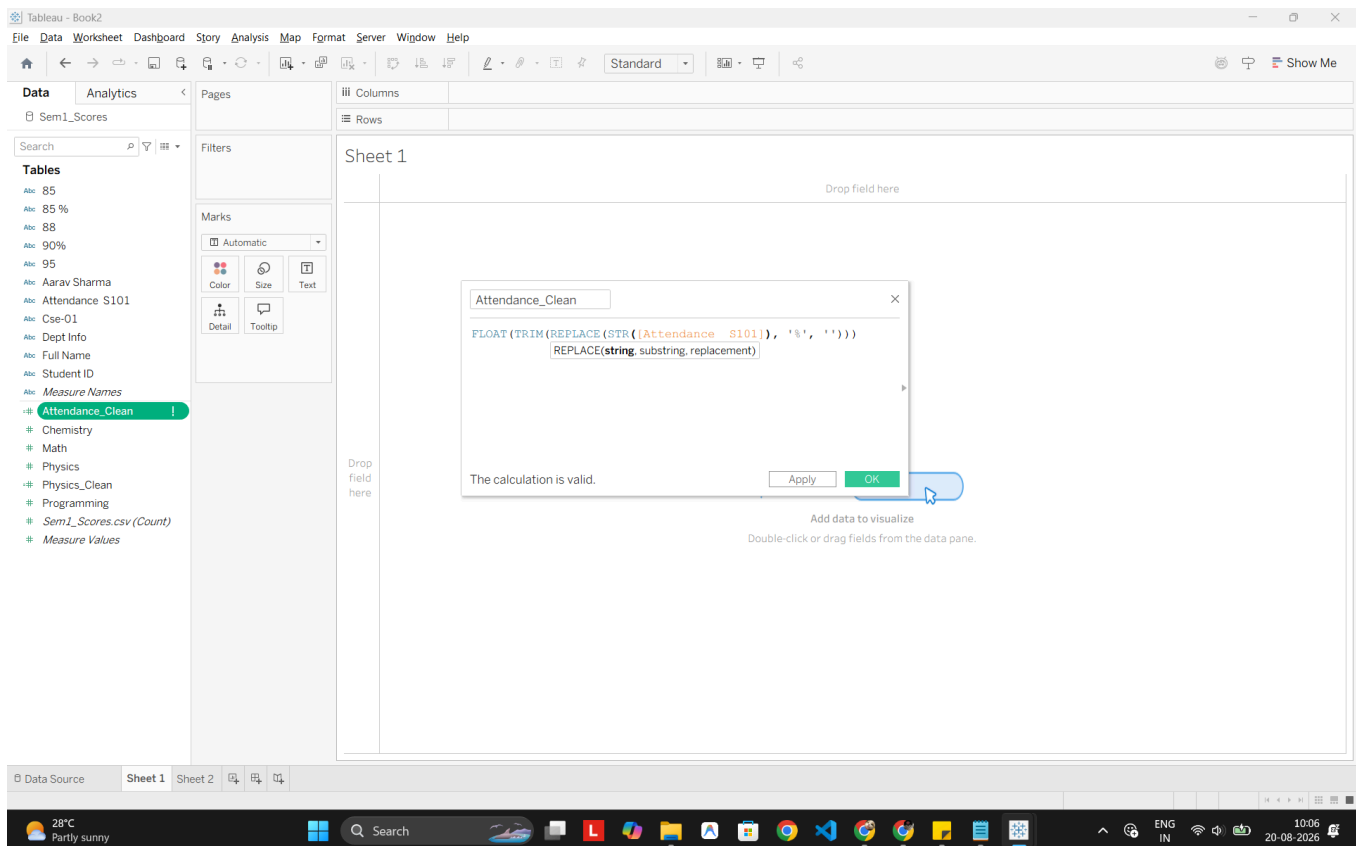
4. Create the cleaned measure for Attendance:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Attendance_Clean
- o Tableau Formula:

Code snippet

FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))

- o Click OK.



Step 3: Handle Missing (Null) Numeric Values via Imputation

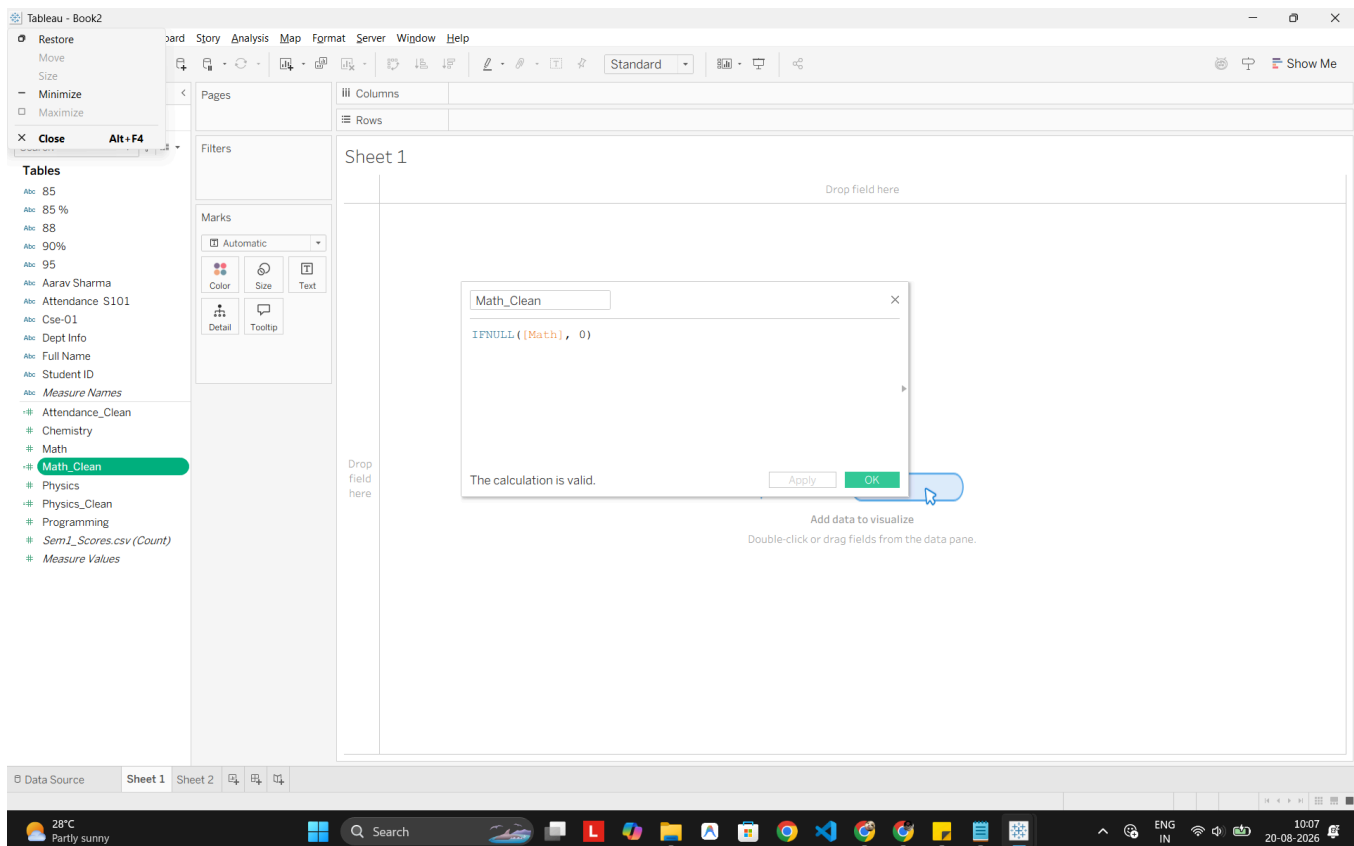
1. Handle missing entries in Math:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Math_Clean
- o Tableau Formula:

Code snippet

```
IFNULL([Math], 0)
```

- o Click OK.



2. Handle missing entries in Chemistry:

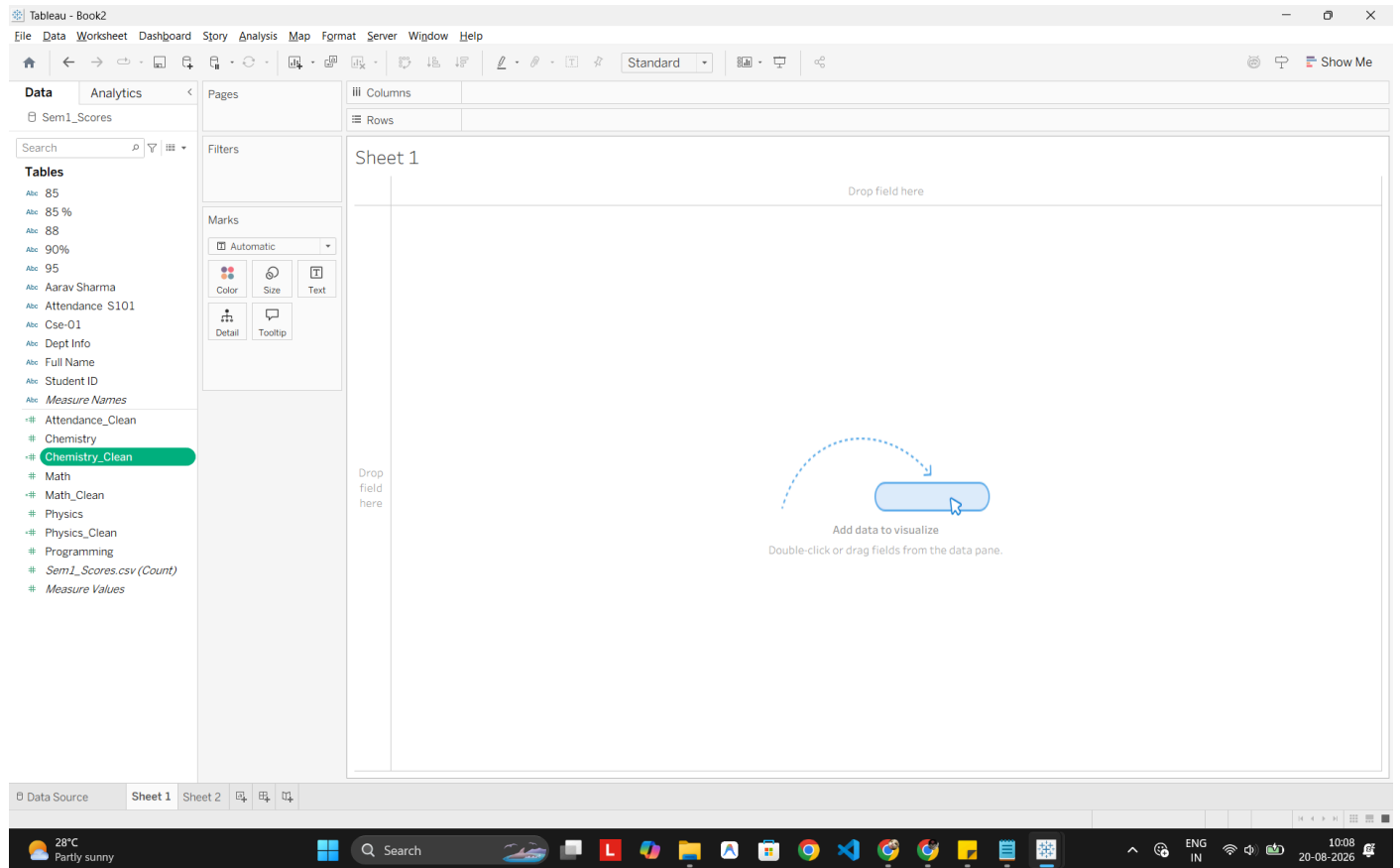
- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Chemistry_Clean
- o Tableau Formula:

Code snippet

```
IFNULL([Chemistry], 0)
```

- o Click OK.

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Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

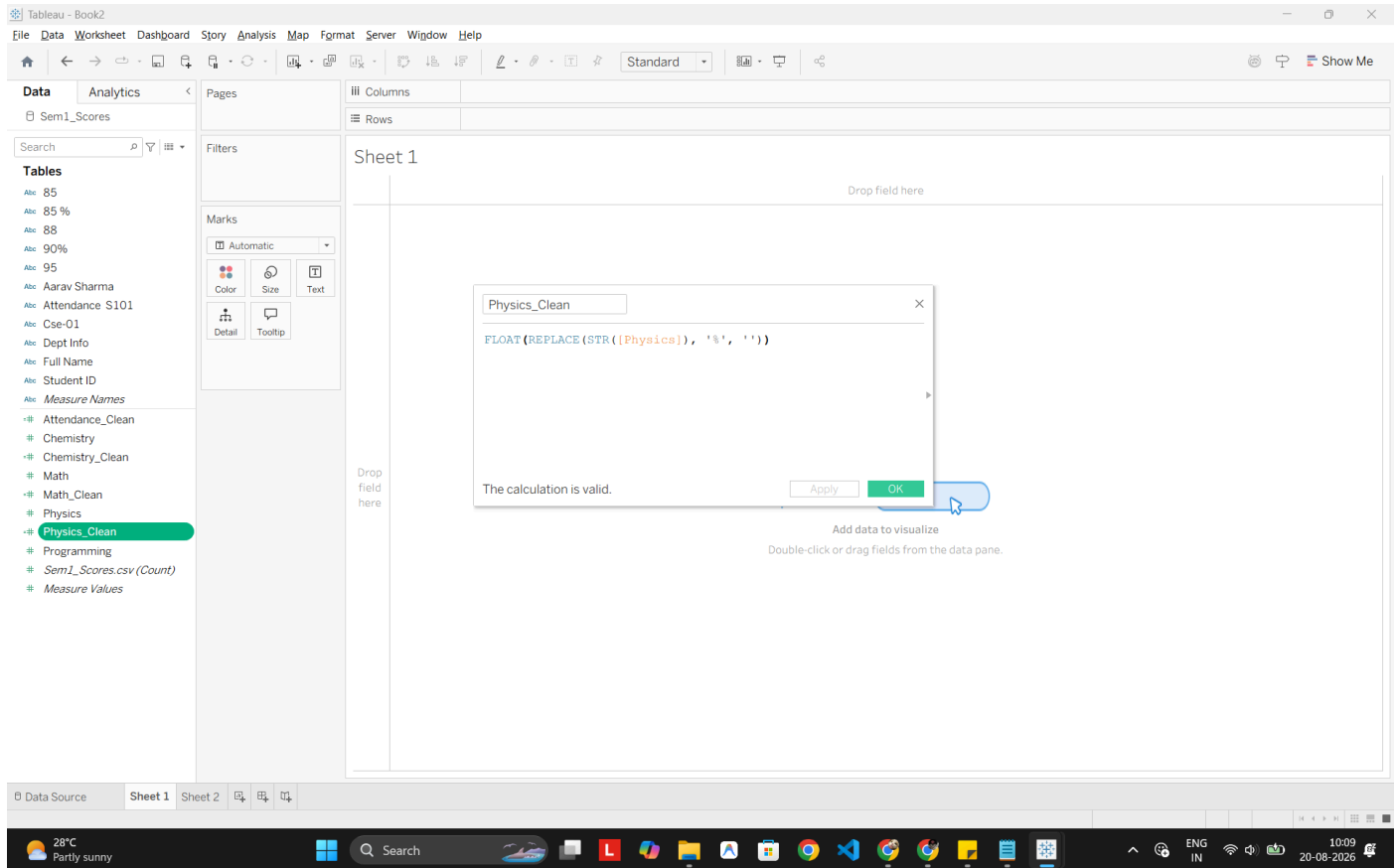
1. Student S105 contains a Null value in Physics. Combine string stripping with null handling:

- o Edit Physics_Clean (right-click Physics_Clean --> Edit...):
- o Updated Tableau Formula:

Code snippet

```
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
```

- o Click OK.



Step 1: Open the Data Source Grid

1. In Tableau, navigate to the Data Source tab at the bottom-left corner.

2. Locate the four horizontal subject score columns:

- o Math
- o Physics
- o Chemistry
- o Programming

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Sem1_Scores' selected. Below it, the 'Files' pane lists several CSV and TXT files. The main workspace displays a diagram with a box labeled 'Sem1_Scores.csv' and two arrows pointing to it, labeled 'Relate tables' and 'Add a base table'. At the bottom, the 'Data Source' tab is active, showing a table with 19 fields and 9 rows. The table has columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, and Programming. The Math, Physics, Chemistry, and Programming columns are highlighted in blue.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming
S102	priya VERMA	ECE-02	null	0.750000	80	85
S103	Rohan Gupta	CSE-01	78	0.820000	null	88
S104	Sneha Patel	IT-03	92	0.880000	90	96
S105	aditya Rao	MECH-04	45	null	50	60
S106	Ananya Singh	CSE-01	88	0.910000	85	92
S107	Rahul Nair	ECE-02	null	0.700000	75	80
S108	Kavya Reddy	IT-03	95	0.940000	92	98
S109	Vikram Joshi	CSE-01	67	0.600000	58	72

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.

2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.

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The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The sidebar on the left contains 'Connections' and 'Files' sections. The main workspace displays a data preview table with columns for Student ID, Full Name, Dept Info, Math, Physics, Chemistry, and Programming. The 'Physics' column is highlighted in blue.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Sem1_Scores.csv
S102	priya VERMA	ECE-02	78	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	92	0.820000	90	88	78 %
S104	Sneha Patel	IT-03	45	0.880000	50	96	95 %
S105	aditya Rao	MECH-04	88	0.910000	85	92	88 %
S106	Ananya Singh	ECE-01	75	0.700000	75	80	72 %
S107	Rahul Nair	ECE-02	95	0.940000	92	98	98 %
S108	Kavya Reddy	IT-03	67	0.600000	58	72	80 %
S109	Vikram Joshi	CSE-01					

3. Right-click on any of the highlighted column headers.
4. Select Pivot.

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Tableau - Book2

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live
Filters: 0

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 19 fields 9 rows

9 rows

Name: Sem1_Scores.csv

Description: No description available.

Fields

Type	Field Name
Math	Math
Physics	Physics
Chemistry	Chemistry
Programming	Programming

Context Menu:

- Rename
- Copy Values
- Hide
- Create Calculated Field...
- Pivot
- Merge Mismatched Fields

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Sem1_Scores.csv
S102	priya VERMA	ECE-02	78	0.750000	80	85	90
S103	Rohan Gupta	CSE-01	92	0.820000	90	88	78
S104	Sneha Patel	IT-03	45	0.880000	50	60	65
S105	aditya Rao	MECH-04	88	0.910000	85	92	88
S106	Ananya Singh	ECE-02	75	0.700000	75	80	72
S107	Rahul Nair	IT-03	95	0.940000	92	98	98
S108	Kavya Reddy	CSE-01	67	0.600000	58	72	80
S109	Vikram Joshi						

Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

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1. Double-click the header named Pivot Field Names and type: Subject
2. Double-click the header named Pivot Field Values and type: Score

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The left sidebar displays the 'Connections' and 'Files' panes. The main workspace shows a data preview for 'Sem1_Scores.csv' with 17 fields and 36 rows. The 'Fields' pane on the left lists the fields: Subject, Score, Physics_Clean, and Attendance_Clea. The data preview table is as follows:

Student ID	Full Name	Dept Info	Attendance S101	Aarav Sharma	Cse-01	Sem1_Scores.csv
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null

Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list.

Tableau - Book2

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Connection: Live Extract
Filters: 0 Add

Sem1_Scores.csv

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores fields 36 rows

Abc Sem1_Scores.csv	Abc Sem1_Scores.csv	Abc Sem1_Scores.csv	Abc Sem1_Scores.csv	Abc Sem1_Scores.csv	Abc Sem1_Scores.csv	Abc Sem1_Scores.csv
Student ID	Full Name	Dept Info	Attendance S101	Aarav Sharma	Cse-01	85
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null

28°C Partly sunny

Search

ENG IN

10:16 20-08-2026

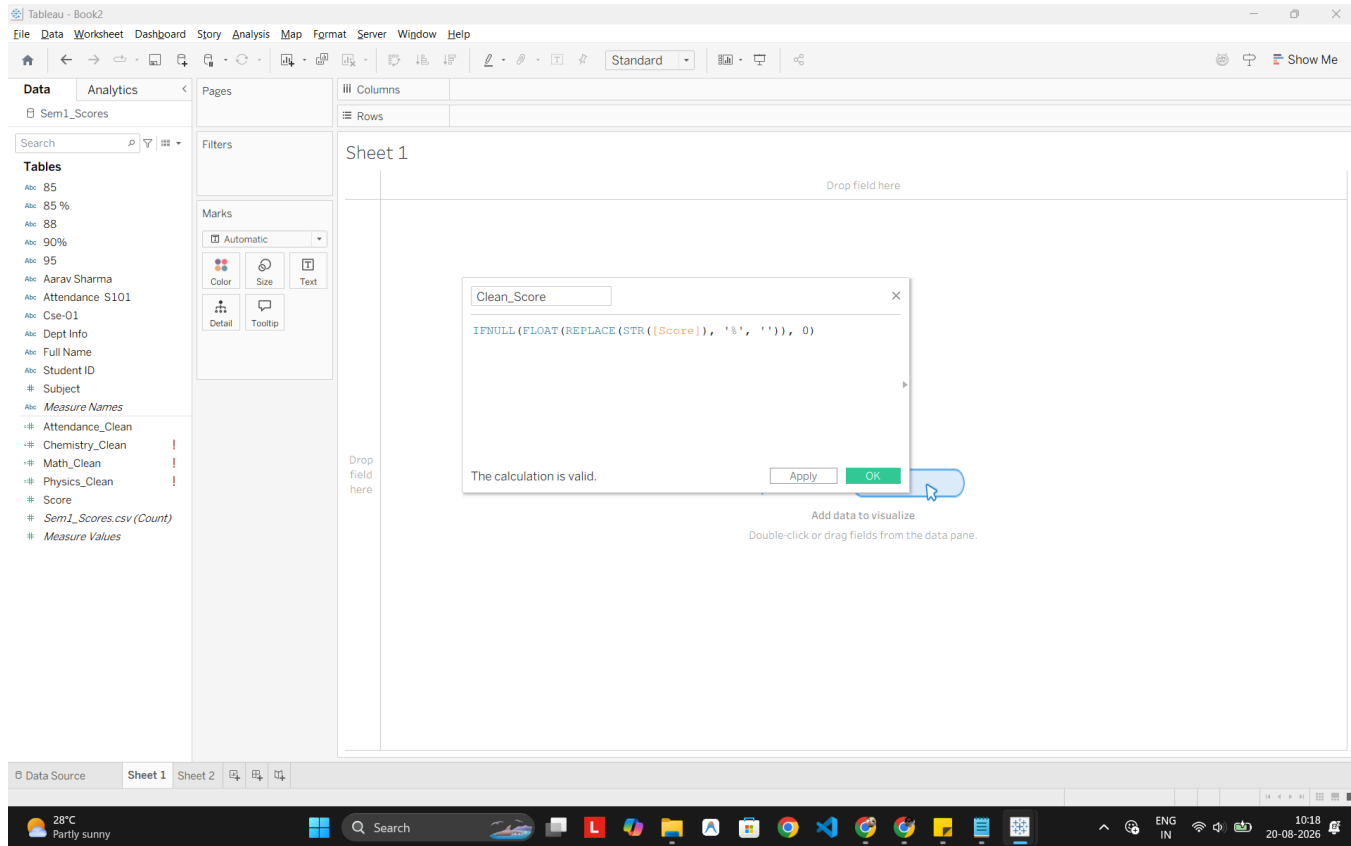
3. Alternatively, create a calculated field inside

Sheet 1 named Clean_Score:

Code snippet

```
IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)
```

C: Perform Union of Multiple Files



Step 1: Open the Data Source Canvas

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Subject: Data Visualization with Power BI and Tableau

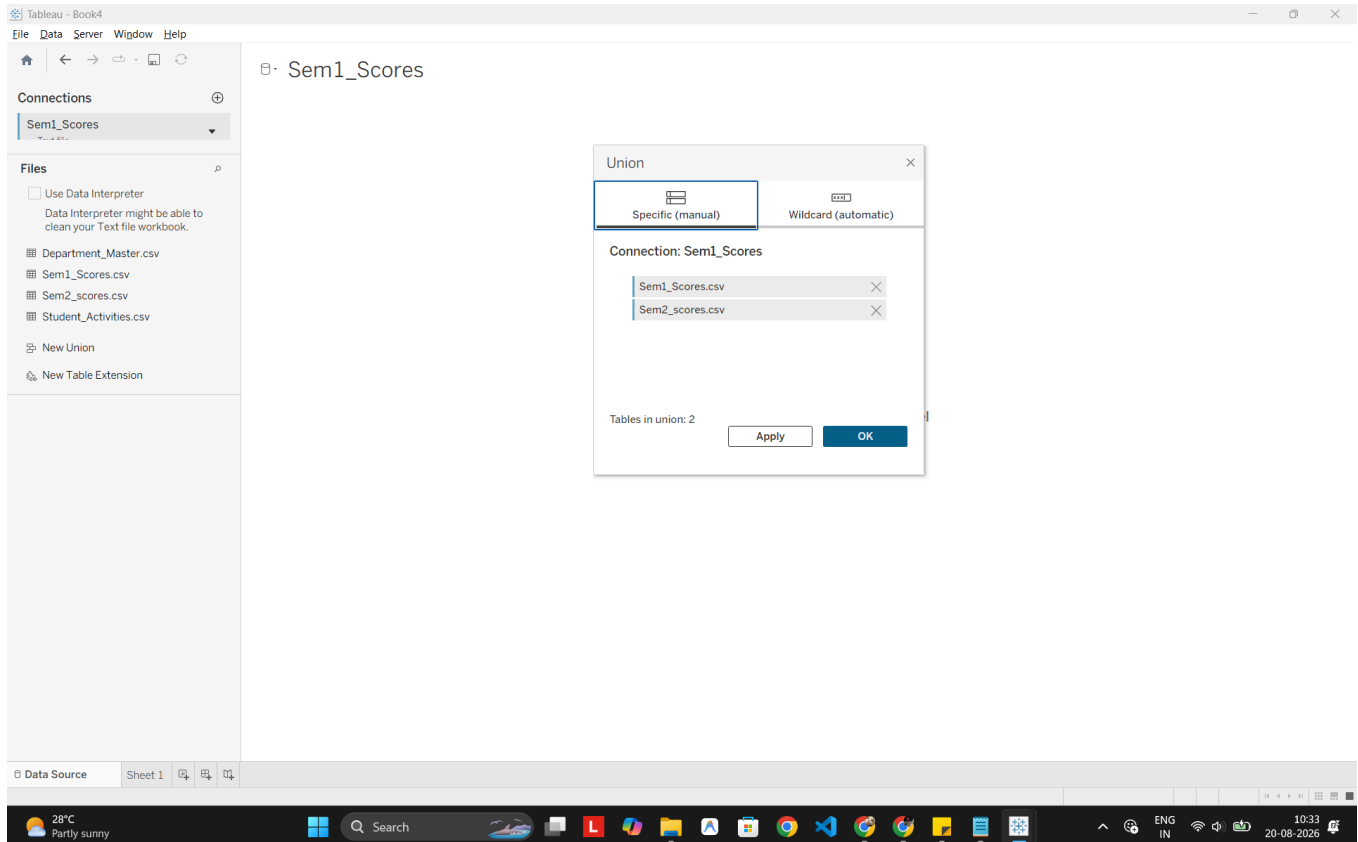
1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as the selected connection. Below it, the 'Files' pane shows a list of files including 'Department_Master.csv', 'Sem1_Scores.csv', 'Sem2_Scores.Csv.txt', and 'Student_Activities.csv'. The main workspace displays a context menu for 'Sem1_Scores.csv' with options: 'Open...', 'Rename', 'Remove' (highlighted), 'Field names are in first row', 'Generate field names automatically', 'Text File Properties...', and 'Convert to Union...'. Below the menu, there are instructions for 'Relate tables' and 'Add a base table'. At the bottom, the 'Data Source' tab is active, showing a table with 18 fields and 36 rows. The table has columns for Student ID, Full Name, Dept Info, Attendance S101, Aarav Sharma, Cse-01, and 85. The 'Fields' section on the left of the table lists fields like Subject, Score, Physics_Clean, and Attendance_Clea with their respective types and physical table names.

Student ID	Full Name	Dept Info	Attendance S101	Aarav Sharma	Cse-01	85
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S102	priya VERMA	ECE-02	90 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78 %	null	null	null

Step 2: Create a Multi-File Union

1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.
3. In the pop-up Union dialog box:
 - o Drag Sem1_Scores.csv from the left panel into the box.
 - o Drag Sem2_Scores.csv from the left panel into the same box.
 - o Verify that both filenames are listed in the Union box.
4. Click Apply, then click OK.



5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.

Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Sem1_Scores' selected. Below it, the 'Files' pane lists several CSV files: 'Department_Master.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', and 'Student_Activities.csv'. The main workspace displays a 'Union' node. Below the node, there are instructions: 'Relate tables' (Drag tables near Union to relate them) and 'Add a base table' (Drag out additional base tables for multi-fact analysis). At the bottom, the 'Data Source' pane shows the 'Union' of two tables, 'Sem1_Scores' and 'Sem2_Scores'. The 'Fields' list on the left of the data grid includes 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', 'Attendance', 'Table Name', and 'Physics'. The data grid itself shows 20 rows of data, with the first 10 rows corresponding to 'Sem1_Scores.csv' and the next 10 rows corresponding to 'Sem2_Scores.csv'. The 'Table Name' column in the data grid shows 'Sem1_Scores.csv' for the first 10 rows and 'Sem2_Scores.csv' for the next 10 rows.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	0.850000	Sem1_Scores.csv
S102	Priya Verma	ECE-02	75	0.750000	80	85	0.900000	Sem1_Scores.csv
S103	Rohan Gupta	CSE-01	78	0.820000	88	95	0.780000	Sem1_Scores.csv
S104	Sneha Patel	IT-03	92	0.880000	90	96	0.950000	Sem1_Scores.csv
S105	Aditya Rao	MECH-04	45	0.450000	50	60	0.650000	Sem1_Scores.csv
S106	Ananya Singh	CSE-01	88	0.910000	85	92	0.880000	Sem1_Scores.csv
S107	Rahul Nair	ECE-02	70	0.700000	75	80	0.720000	Sem1_Scores.csv
S108	Kavya Reddy	IT-03	95	0.940000	92	98	0.980000	Sem1_Scores.csv
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	0.800000	Sem1_Scores.csv
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	0.910000	Sem2_Scores.csv

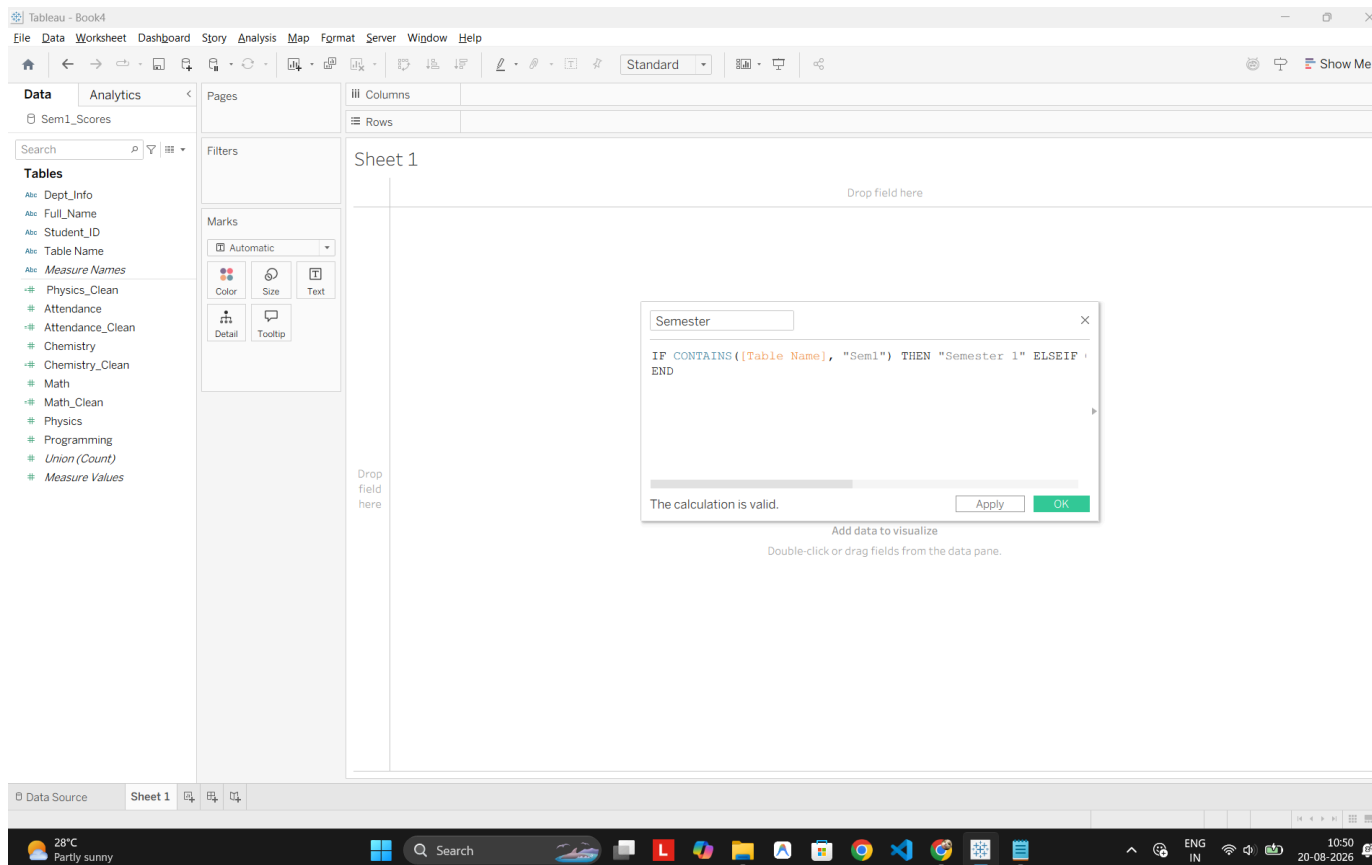
3. Scroll to the far right of the grid to verify the generated system column:
 - o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space and select Create Calculated Field....
3. Set the Field Name to: Semester 4. Enter the following formula:

Code snippet

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"  
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"  
ELSE "Other"  
END
```

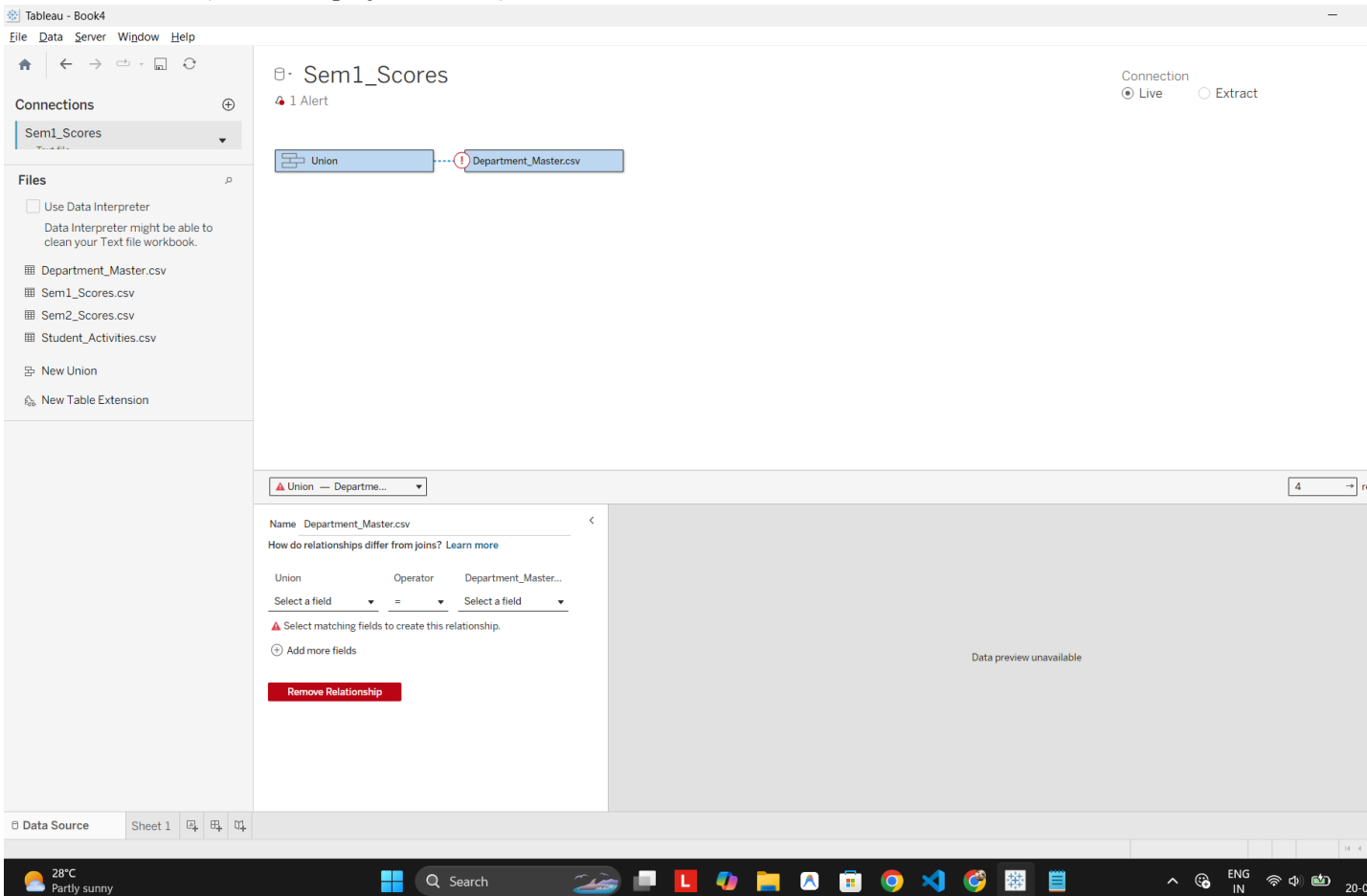


Step 1: Stay on the Top Canvas

🏠 Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Unionbox.
3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):
- o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

Tableau - Book4

FileDataServerWindowHelp

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

Sem1_Scores.csv

Sem2_Scores.csv

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Connection

LiveExtract

Filters

0Add

Union

Department_Master.csv

Union --- Departme...

4 rows

NameDepartment_Master.csv

How do relationships differ from joins? Learn more

Union

Operator

Department_Master...

Dept_Info = Dept_Code

Add more fields

Performance Options

Remove Relationship

Department_Master.csv

Department_Master.csv

Department_Master.csv

Department_Master.csv

Department_Master.csv

Dept Code

Department Name

HOOD Name

Dept Target Average

Total Dept Capacity

CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Data Source

Sheet 1

10:54 20-08-2026

Step 3: Add and Link Student_Activities.csv

- 1. From the left sidebar, drag Student_Activities.csv onto the canvas.
- 2. Drop it to the right of the Union box.

Tableau - Book4

FileDataServerWindowHelp

HomePreviousNextRefresh

Connections

Sem1_Scores

Files

☐ Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

Sem1_Scores.csv

Sem2_Scores.csv

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Connection

LiveExtract

Filters

0Add

Union

Department_Master.csv

Student_Activities.csv

Union -- Student_A...

10 rows

NameStudent_Activities.csv

How do relationships differ from joins? Learn more

Union

Operator

Student_Activities...

Student_ID

=

Student ID (Stu...

Add more fields

Performance Options

Remove Relationship

Student_Activities.csv

Student ID (Student Activi

Student_Activities.csv

Club Membership

Student_Activities.csv

Projects Completed

Student_Activities.csv

Certifications

S101

Coding Club

3

2

S102

Robotics Club

2

1

S103

Coding Club

2

1

S104

Debate Society

4

3

S105

Sports Club

1

0

S106

Coding Club

3

2

S107

Robotics Club

2

1

S108

Debate Society

4

3

S109

Sports Club

1

1

S110

Cultural Club

2

2

28°C

Partly sunny

Search

ENG

IN

10:56

20-08-2026

3. In the relationship dialog:
- o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID
 - o Operator: =

Screenshot of Tableau Desktop interface showing a Union relationship between Department_Master.csv and Student_Activities.csv.

Connections: Sem1_Scores

Files: Department_Master.csv, Sem1_Scores.csv, Sem2_Scores.csv, Student_Activities.csv, New Union, New Table Extension

Relationship Diagram: Union (Left) connected to Department_Master.csv and Student_Activities.csv (Right).

Relationship Configuration:

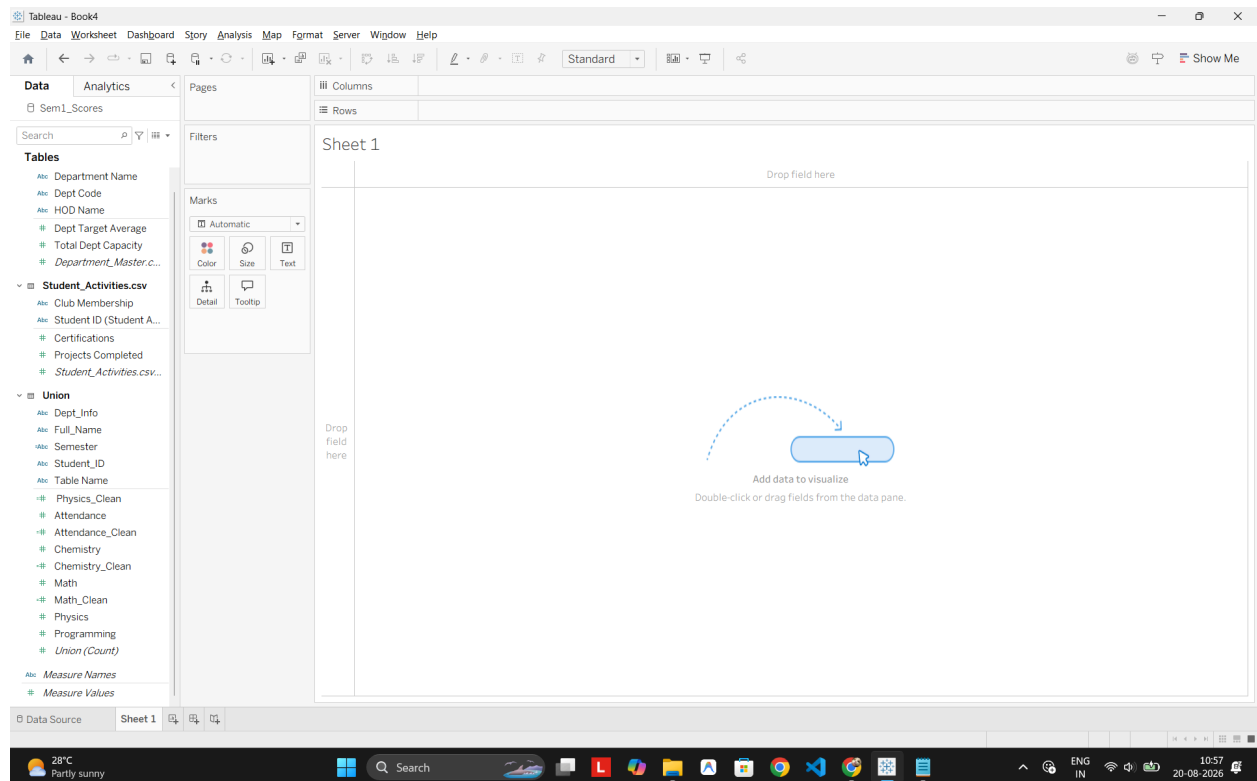
- Name: Student_Activities.csv
- How do relationships differ from joins? [Learn more](#)
- Left Table Field (Union): Student_ID
- Operator: =
- Right Table Field (Student_Activities): Student ID (Stu...
- Buttons: Add more fields, Performance Options, Remove Relationship

Data Table:

Student ID (Student Activ	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1
S108	Debate Society	4	3
S109	Sports Club	1	1
S110	Cultural Club	2	2

Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications



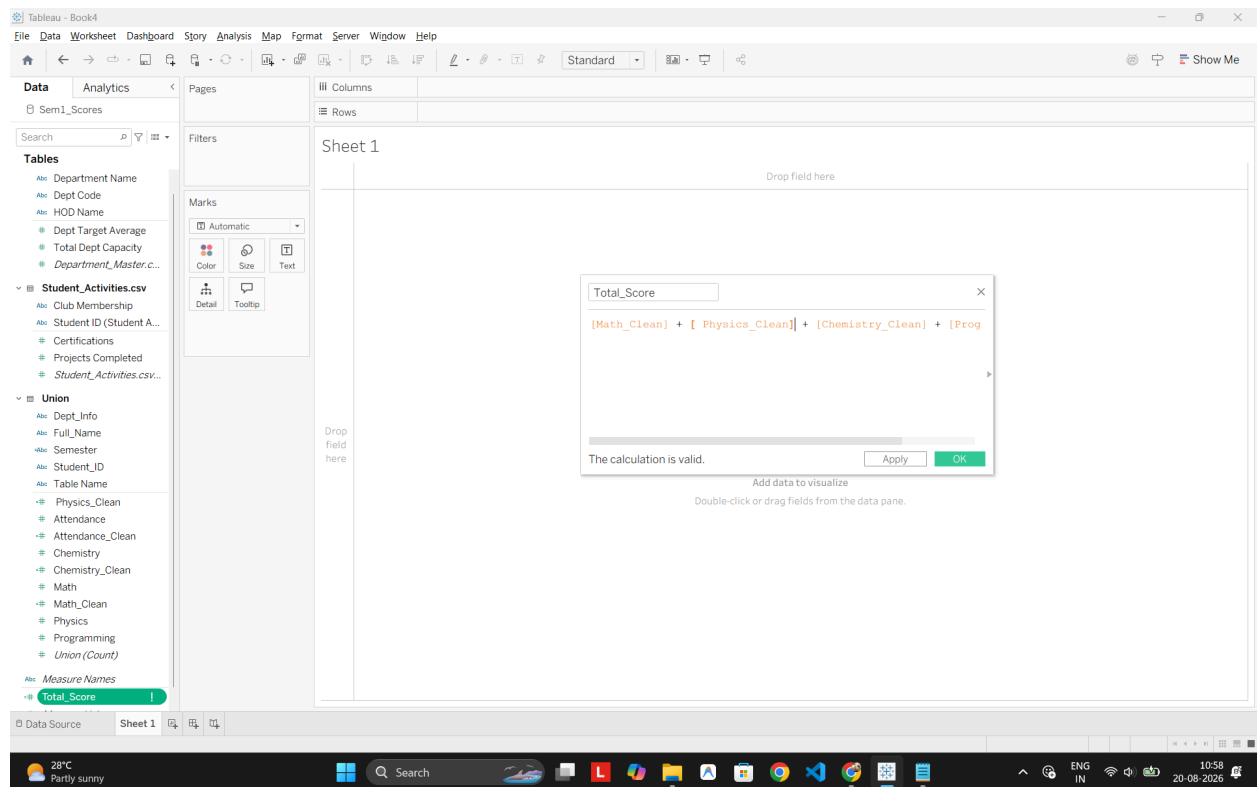
Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the student's total marks field:
 - o Field Name: Total_Score
 - o Tableau Formula:

Code snippet

[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] +
[Programming]

- o Click OK.



4. Create the scaled overall percentage field:

- o Right-click in the Data Pane --> select Create Calculated Field....

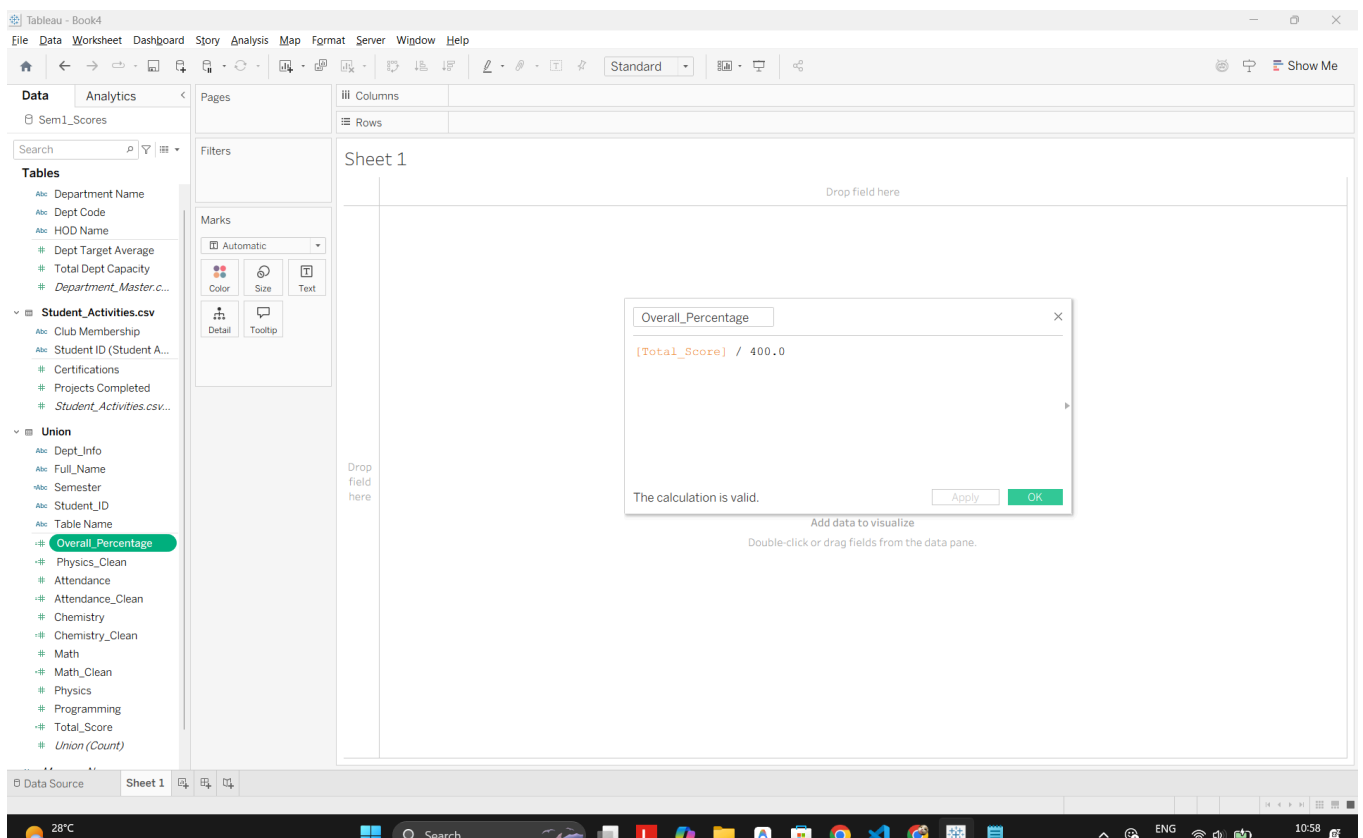
- o Field Name: Overall_Percentage

- o Tableau Formula:

Code snippet

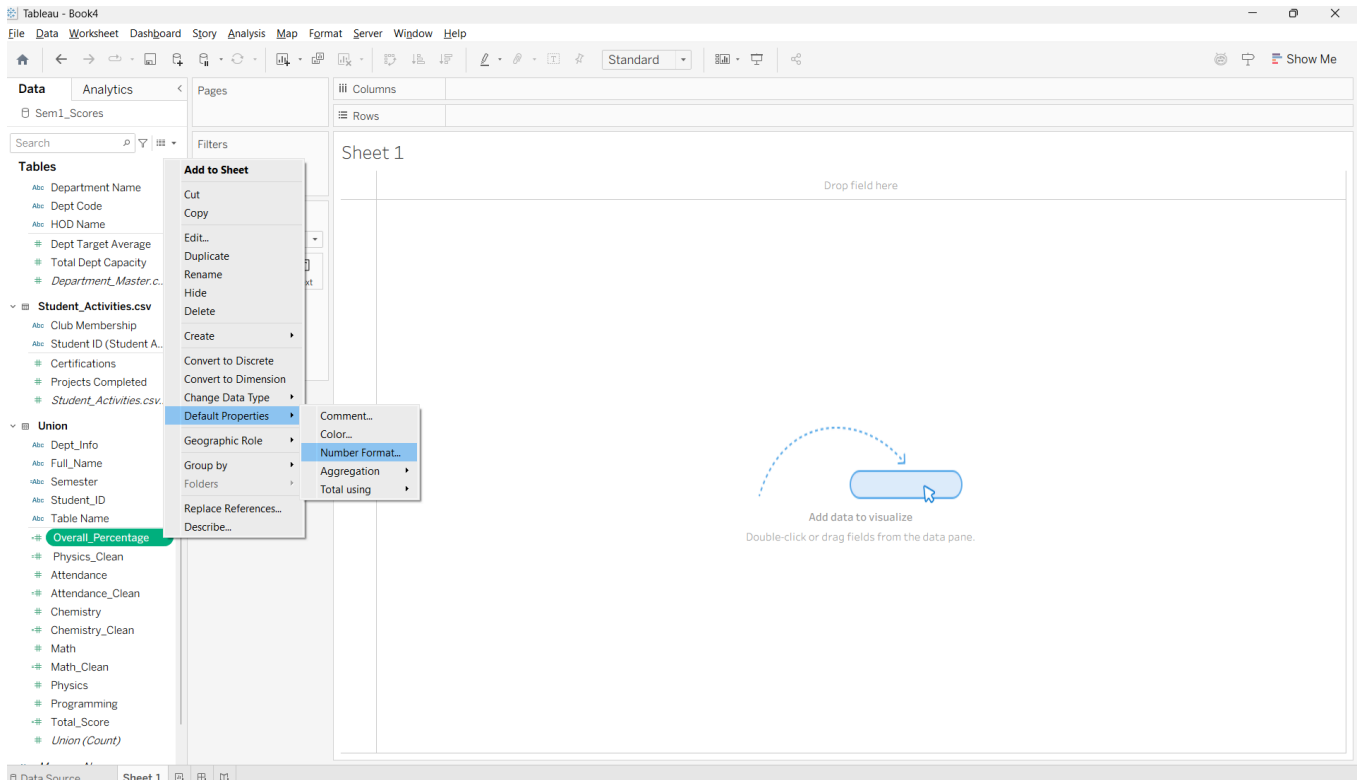
$[Total_Score] / 400.0$

- o Click OK.

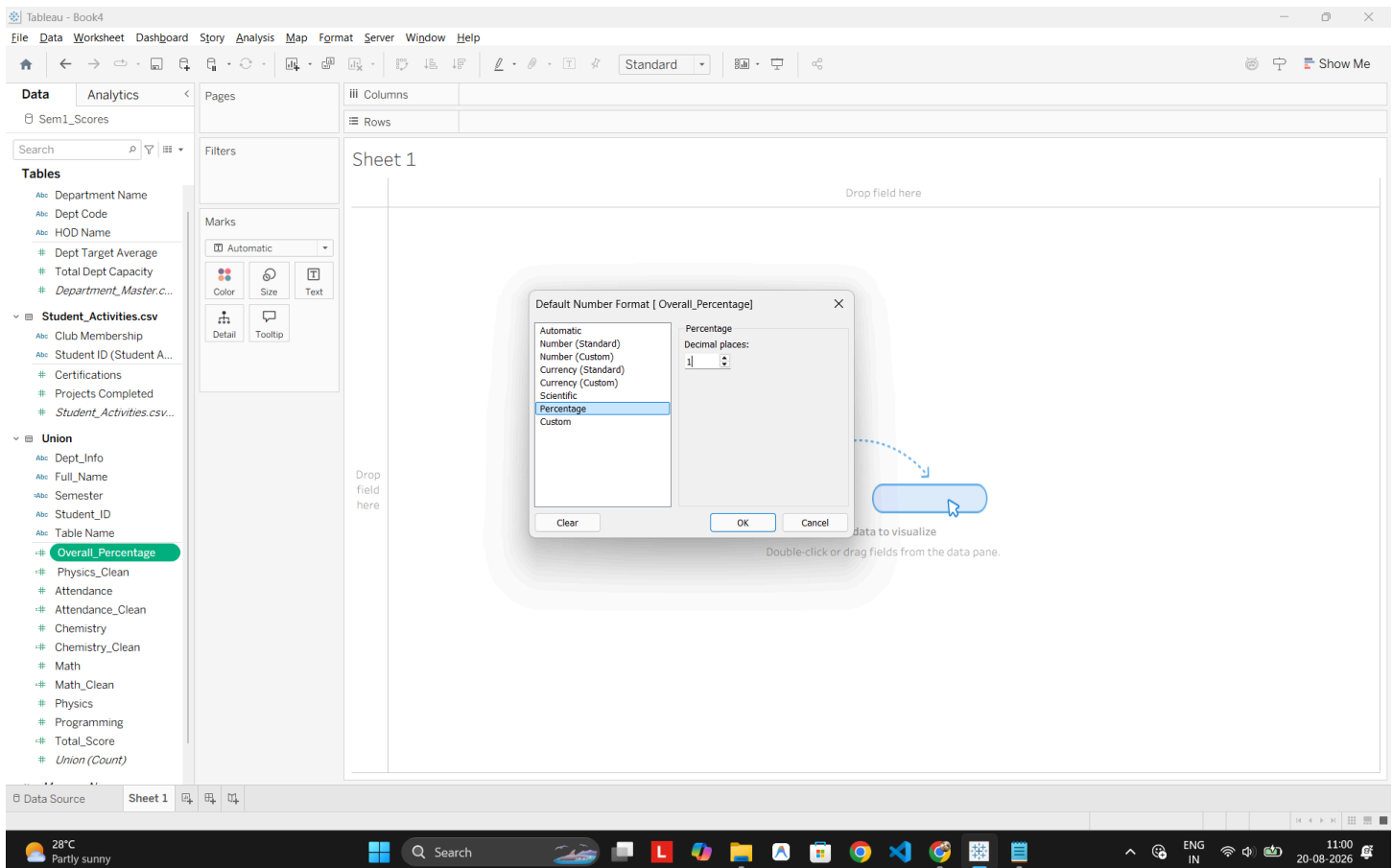


5. Set the default percentage display formatting:

- o In the Data Pane, right-click Overall_Percentage -->Default Properties -->Number Format....



- o Select Percentage and set decimal places to 1 (e.g., 85.5%).
- o Click OK.

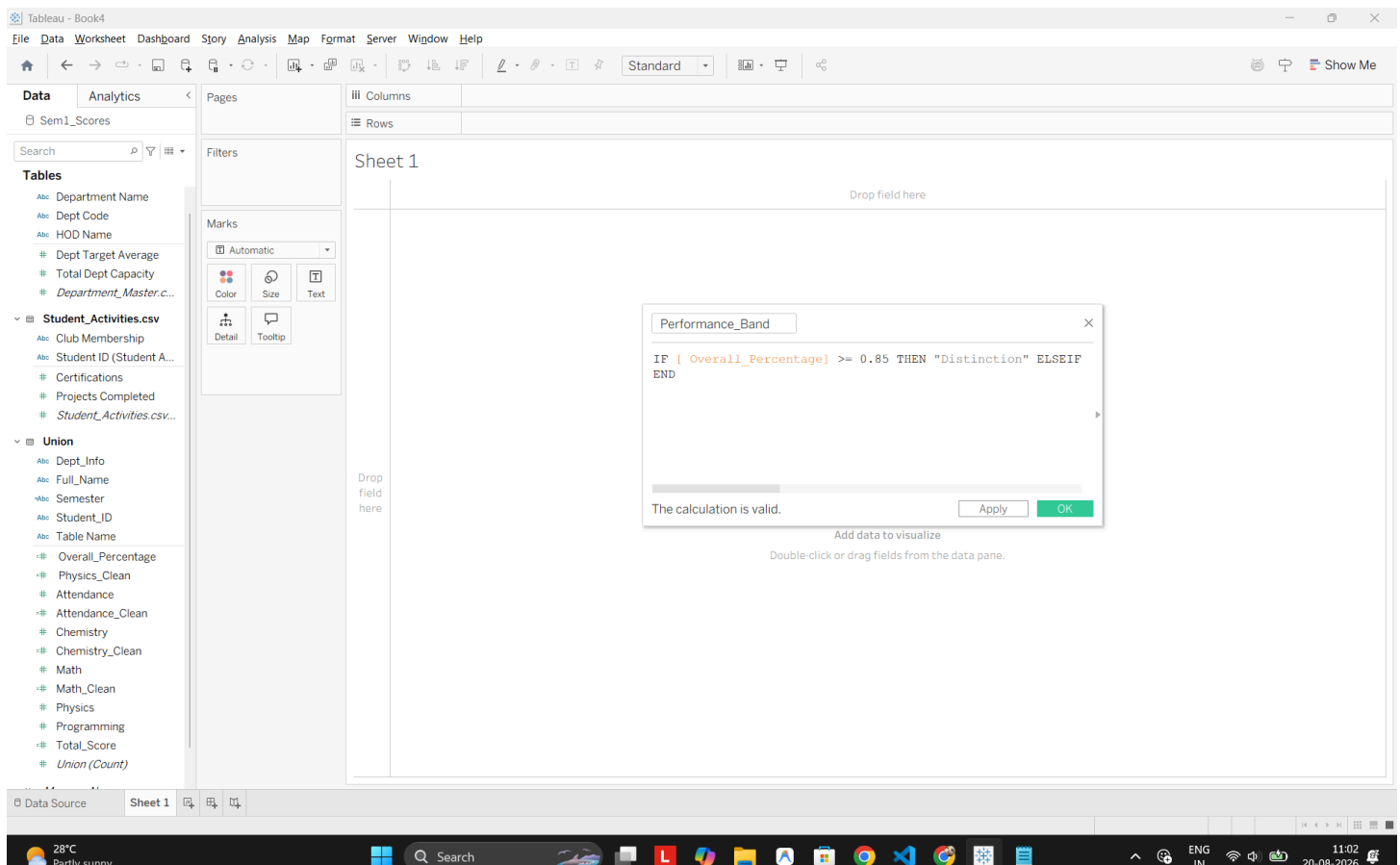


Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the academic grading tier field:
 - o Field Name: Performance_Band
 - o Tableau Formula:

Code snippet

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"  
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"  
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"  
ELSE "Needs Academic Support" END
```



Step

3: Create an Attendance Health Flag

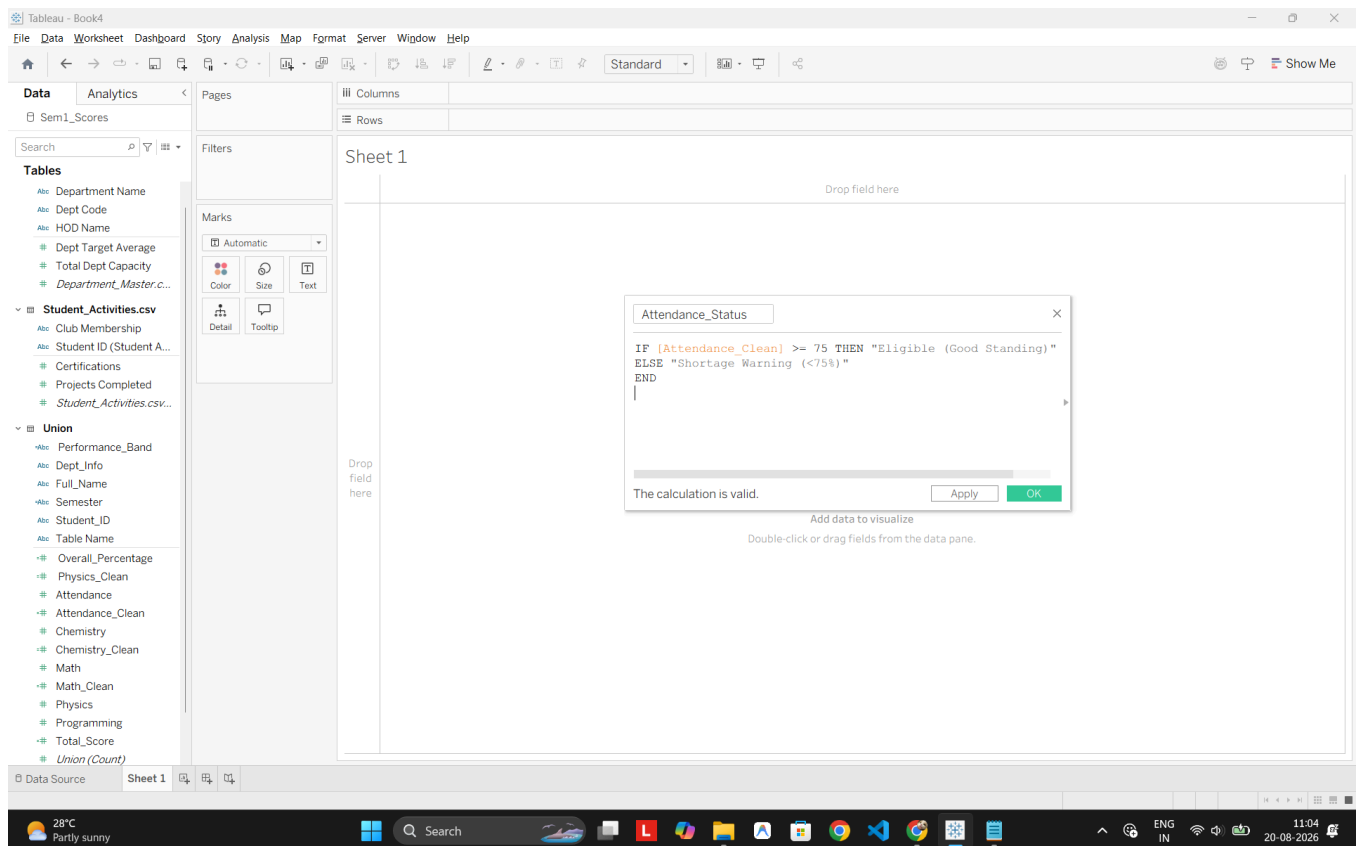
1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the attendance compliance indicator:
 - o Field Name: Attendance_Status
 - o Tableau Formula:

Code snippet

```
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"  
ELSE "Shortage Warning (<75%)"  
END
```

- o Click OK.

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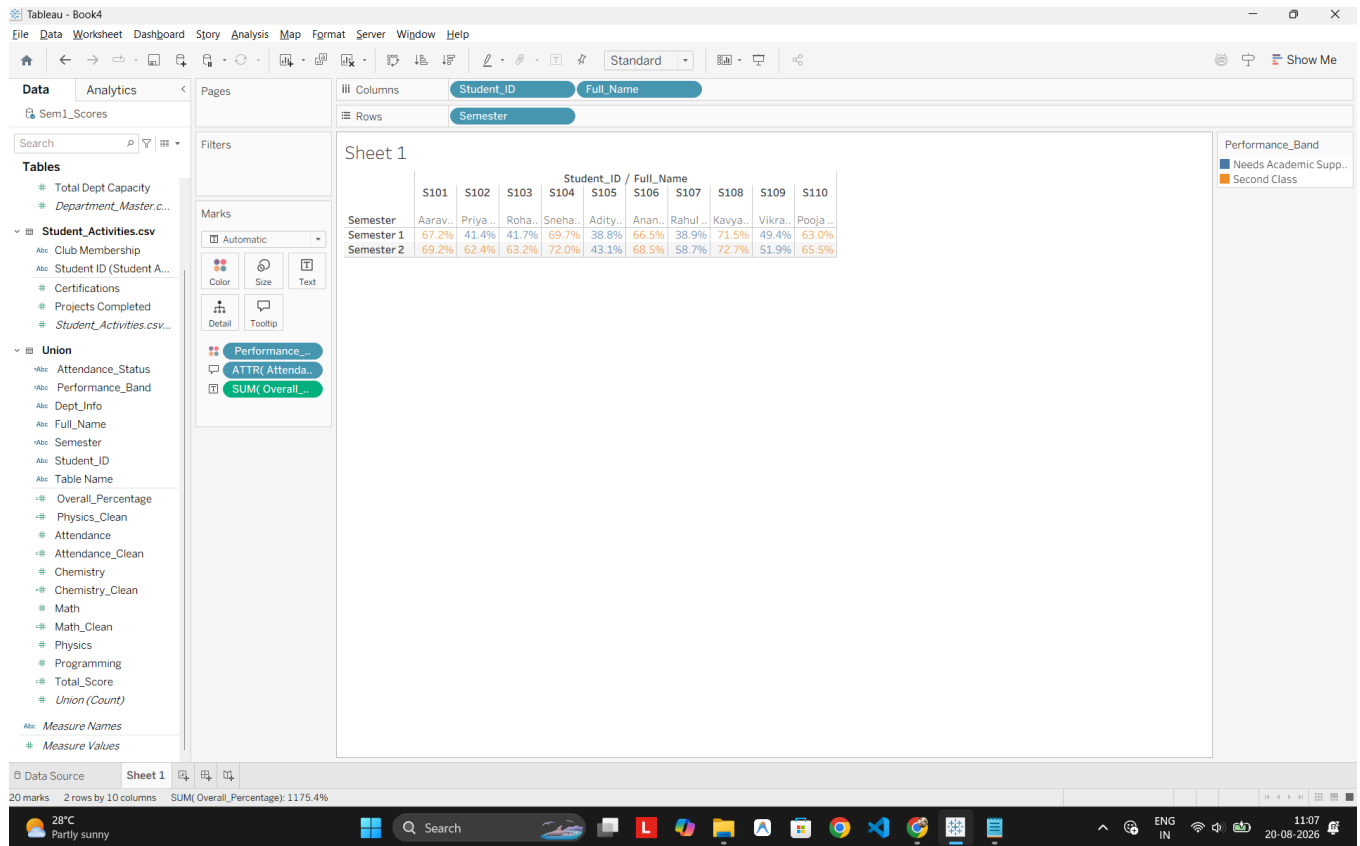


Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.

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Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2). 2. In the Data Pane on

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the left, right-click an empty space --> select Create Calculated Field....

3. Create the department-wide benchmark field:

o Field Name: Dept_Average_Percentage_LOD o Tableau Formula:

Code snippet

```
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }
```

o Click OK.

The screenshot shows the Tableau Desktop interface. On the left, the 'Data' pane lists various data sources and fields. The 'Marks' card is set to 'Automatic'. The 'Columns' shelf contains 'Student_ID' and 'Full_Name', and the 'Rows' shelf contains 'Semester'. The main view displays a table with student performance data across two semesters. A dialog box titled 'Average_Percentage_LOD' is open, showing the calculated field formula: `{ FIXED (Dept_Info) : AVG ( Overall_Percentage ) }`. The dialog has 'Apply' and 'OK' buttons.

	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110
Semester	Aarav...	Priya...	Roha...	Sneha...	Adity...	Anan...	Rahul...	Kavya...	Vikra...	Pooja...
Semester 1	67.2%	41.4%	41.7%	69.7%	38.8%	66.5%	38.9%	71.5%	49.4%	63.0%
Semester 2	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%

. Set default formatting:

- o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.

The screenshot shows the Tableau Desktop interface. The main view is a data table titled 'Sheet 1' with columns for 'Semester', 'Student_ID', 'Full_Name', and 'Dept_Average_Percentage_LOD'. The table displays data for two semesters across 10 students. A dialog box titled 'Default Number Format [Dept_Average_Percentage_LOD]' is open, showing the 'Percentage' format selected with 'Decimal places' set to 1. Another dialog box is partially visible in the background.

	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110
Semester	Aarav...	Priya...	Roha...	Sneha...	Adity...	Anan...	Rahul...	Kavya...	Vikra...	Pooja...
Semester 1	67.2%	41.4%	41.7%	69.7%	38.8%	66.5%	38.9%	71.5%	49.4%	63.0%
Semester 2	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%

Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the cumulative student benchmark across all semesters:
 - o Field Name: Student_Overall_Career_Average
 - o Tableau Formula: Code snippet

{ FIXED [Student_ID] : AVG([Overall_Percentage]) }

o Click OK.

The screenshot shows the Tableau Desktop interface. On the left, the Data Pane lists several tables, including 'Student_Activities.csv' and 'Union'. The main view displays a table with columns for 'Semester', 'Student_ID', 'Full_Name', and 'Overall_Percentage'. A dialog box is open for creating a calculated field named 'Overall_Career_Average'. The formula entered is: { FIXED [Student_ID] : AVG([Overall_Percentage]) }. The status bar at the bottom indicates '20 marks' and '2 rows by 10 columns'.

Semester	Student_ID	Full_Name	Overall_Percentage
Semester 1	Aarav	Priya	67.2%
Semester 1	Roha	Sneha	41.4%
Semester 1	Aditya	Anan	41.7%
Semester 1	Rahul	Kavya	69.7%
Semester 1	Vikra	Pooja	38.8%
Semester 2	Aarav	Priya	66.5%
Semester 2	Roha	Sneha	38.9%
Semester 2	Aditya	Anan	71.5%
Semester 2	Rahul	Kavya	49.4%
Semester 2	Vikra	Pooja	63.0%

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the relative difference metric:

o Field Name: Variance_From_Dept_Average o Tableau Formula:

Code snippet

$$[\text{Overall_Percentage}] - [\text{Dept_Average_Percentage_LOD}]$$

o Click OK.

The screenshot shows the Tableau Desktop interface. On the left, the Data pane lists various fields, including 'Overall_Percentage' and 'Performance_Band'. The main view displays a table with columns for 'Semester', 'Student_ID', and 'Full_Name'. The 'Overall_Percentage' field is highlighted in green, and the 'Performance_Band' field is highlighted in blue. A dialog box titled 'Variance_From_Dept_Average' is open, showing the formula:
$$[\text{Overall_Percentage}] - [\text{Dept_Average_Percentage_LOD}]$$
 The dialog box has 'Apply' and 'OK' buttons. The table data is as follows:

Semester	Student_ID	Full_Name	Overall_Percentage	Performance_Band
Semester 1	S101	Aarav...	67.2%	Needs Academic Supp..
Semester 1	S102	Priya...	41.4%	Needs Academic Supp..
Semester 1	S103	Roha...	41.7%	Needs Academic Supp..
Semester 1	S104	Sneha...	69.7%	Second Class
Semester 1	S105	Adity...	38.8%	Needs Academic Supp..
Semester 1	S106	Anan...	66.5%	Second Class
Semester 1	S107	Rahul...	38.9%	Needs Academic Supp..
Semester 1	S108	Kavya...	71.5%	Second Class
Semester 1	S109	Vikra...	49.4%	Needs Academic Supp..
Semester 1	S110	Pooja...	63.0%	Second Class
Semester 2	S101	Aarav...	69.2%	Second Class
Semester 2	S102	Priya...	62.4%	Second Class
Semester 2	S103	Roha...	63.2%	Second Class
Semester 2	S104	Sneha...	72.0%	Second Class
Semester 2	S105	Adity...	43.1%	Needs Academic Supp..
Semester 2	S106	Anan...	68.5%	Second Class
Semester 2	S107	Rahul...	58.7%	Needs Academic Supp..
Semester 2	S108	Kavya...	72.7%	Second Class
Semester 2	S109	Vikra...	51.9%	Needs Academic Supp..
Semester 2	S110	Pooja...	65.5%	Second Class

3. Set its Number Format to Percentage (1 decimal place).

Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create an automated comparison label:
 - o Field Name: Benchmark_Status
 - o Tableau Formula:

Code snippet

```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"  
ELSE "Below Department Benchmark" END
```

- o Click OK.

The screenshot shows the Tableau Desktop interface. On the left, the 'Data' pane lists various tables and fields. The 'Marks' card is set to 'Automatic'. The 'Columns' shelf contains 'Student_ID' and 'Full_Name', and the 'Rows' shelf contains 'Semester'. A dialog box titled 'Benchmark_Status' is open, displaying the following formula:

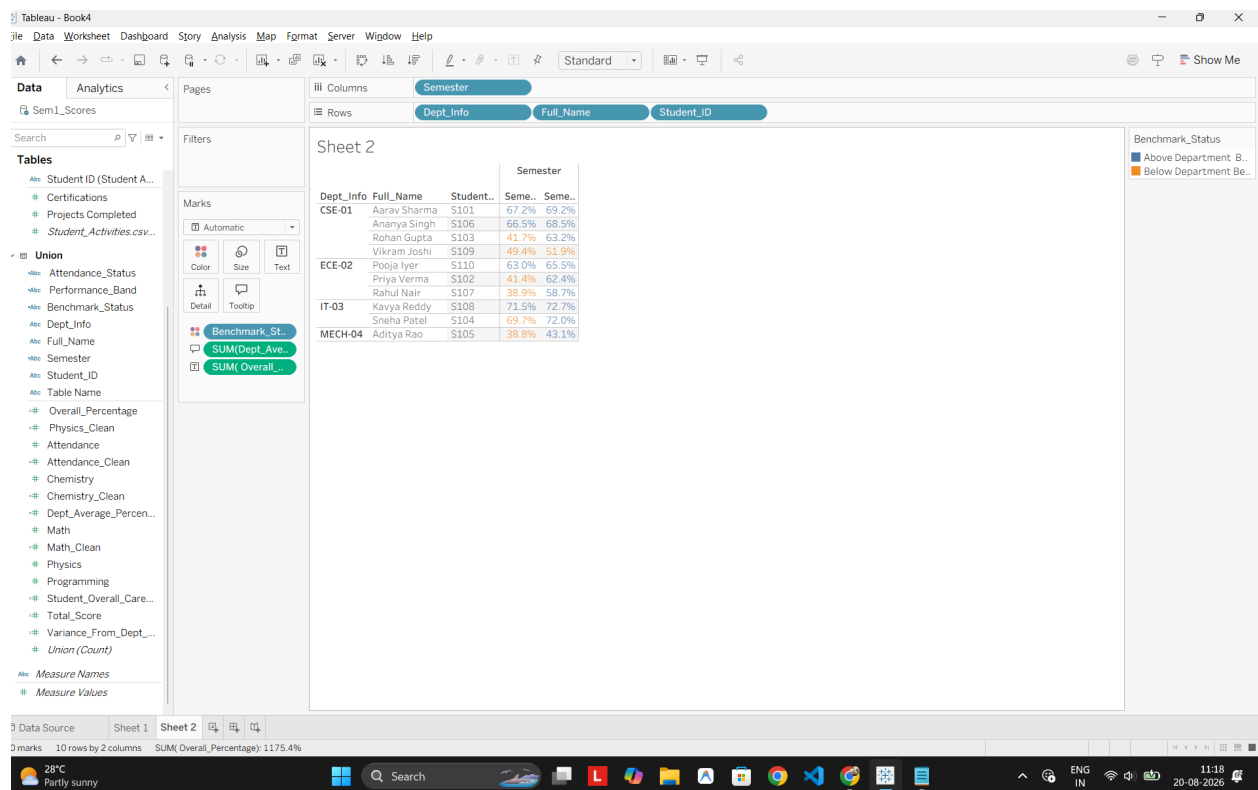
```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"  
ELSE "Below Department Benchmark" END
```

The dialog box has 'Apply' and 'OK' buttons at the bottom right. In the background, a table view is visible with columns for 'Semester', 'Student_ID', and 'Full_Name'. The data is organized by semester (Semester 1 and Semester 2) and lists students with their respective scores.

Semester	Student_ID	Full_Name	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110
Semester 1	Aarav...	Priya...	67.2%	41.4%	41.7%	69.7%	38.8%	66.5%	38.9%	71.5%	49.4%	63.0%
Semester 2	Aarav...	Priya...	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%

Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.



Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:

- o Field Name: Full_Name_Clean
- o Tableau Formula:

Code snippet

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) +  
" " +  
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) +  
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,  
LEN(TRIM([Full_Name]))))
```

- o (Alternatively, for standard trim-only cleaning):

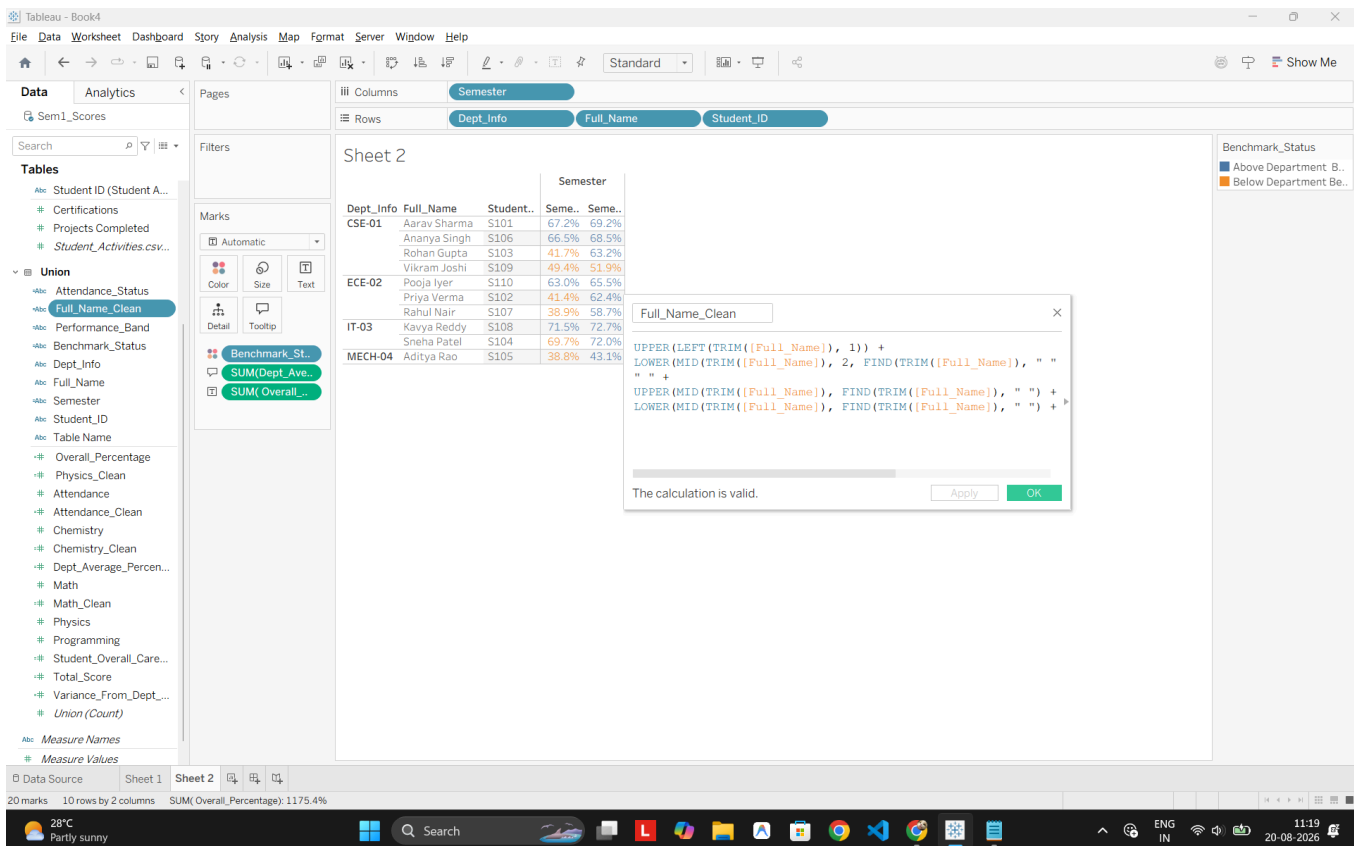
Code snippet

```
TRIM([Full_Name])
```

- o Click OK.

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Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: First_Name
- o Tableau Formula:

Code snippet

```
TRIM(SPLIT([Full_Name], " ", 1))
```

- o Click OK.

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The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Semester'. The 'Rows' shelf contains 'Dept_Info', 'Full_Name', and 'Student_ID'. The 'Marks' shelf is set to 'Automatic'. A data table is displayed on 'Sheet 2' with columns: Dept_Info, Full_Name, Student_ID, and Semester. The table contains data for four departments: CSE-01, ECE-02, IT-03, and MECH-04, each with five students and their respective scores. A dialog box titled 'First_Name' is open, showing the formula `TRIM(SPLIT([Full_Name], " ", 1))`. The status bar at the bottom indicates '20 marks' and '10 rows by 2 columns'.

Dept_Info	Full_Name	Student_ID	Semester
CSE-01	Aarav Sharma	S101	67.2%
	Ananya Singh	S106	66.5%
	Rohan Gupta	S103	41.7%
	Vikram Joshi	S109	49.4%
	Pooja Iyer	S110	63.0%
ECE-02	Priya Verma	S102	41.4%
	Rahul Nair	S107	38.9%
	Kavya Reddy	S108	71.5%
	Sneha Patel	S104	69.7%
	Aditya Rao	S105	38.8%
IT-03			
MECH-04			

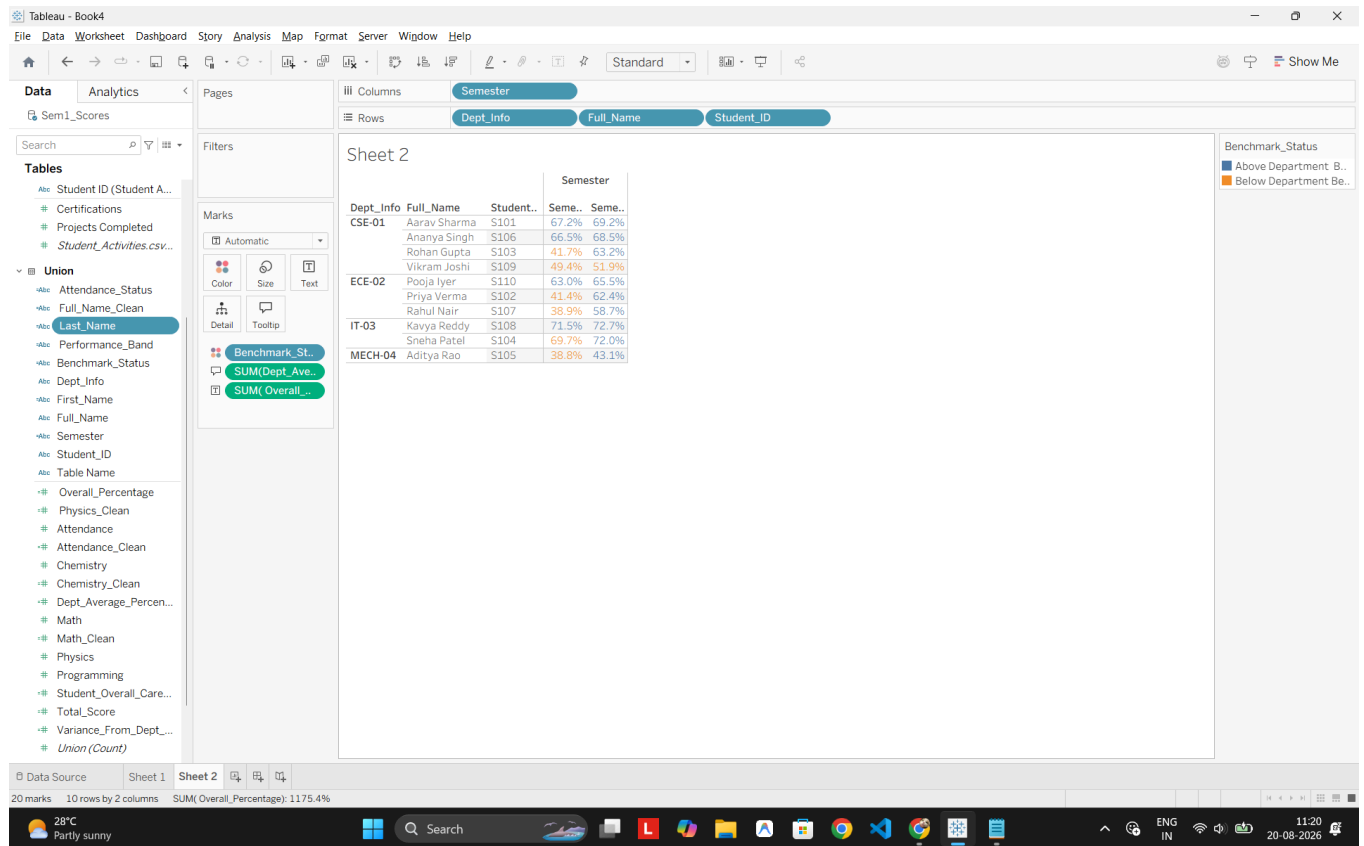
2. Extract the Last Name:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Last_Name
- o Tableau Formula:

Code snippet

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TRIM(SPLIT([Full_Name], " ", -1)) o Click OK.



Step 3: Split Composite Department Codes (Dept_Info)

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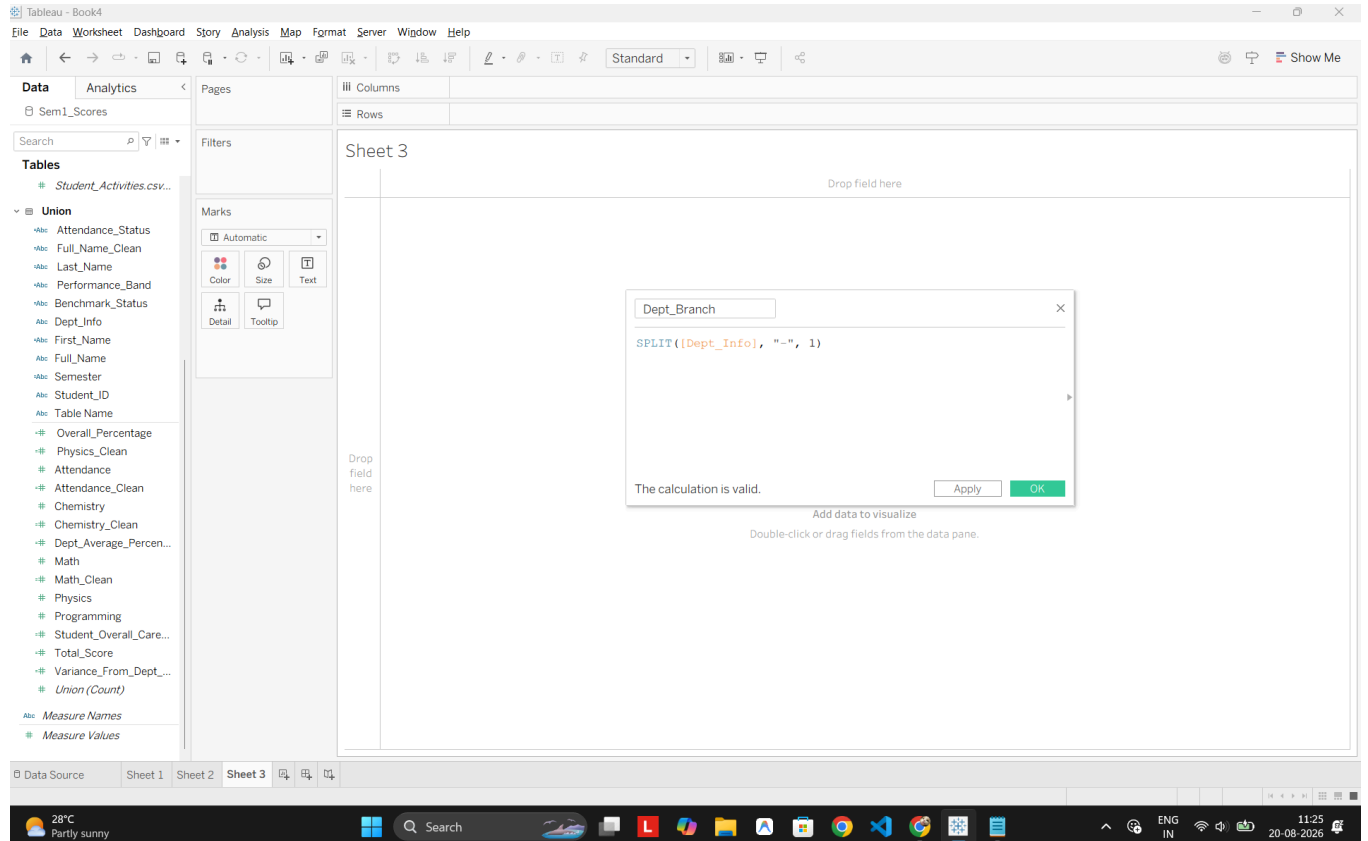
The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04.
Separate them into distinct dimensional attributes:

1. Extract the Department Acronym:

- o Right-click in the Data Pane --> select Create Calculated Field...
- o Field Name: Dept_Branch
- o Tableau Formula:

Code snippet

`SPLIT([Dept_Info], "-", 1)`



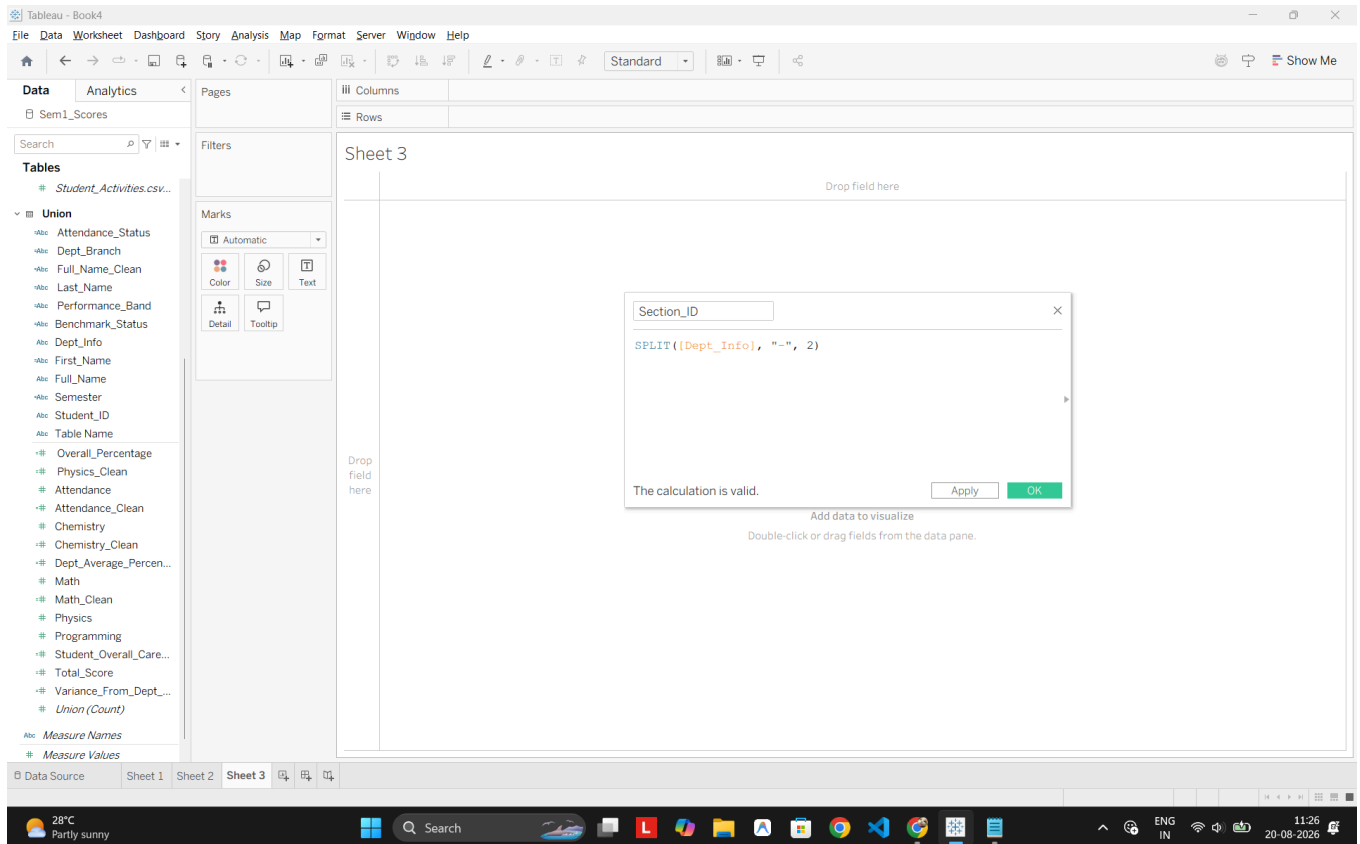
- o Click OK.

2. Extract the Batch / Section ID:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Section_ID
- o Tableau Formula:

Code snippet

SPLIT([Dept_Info], "-", 2)



- o Click OK.

Step 4: Create a Clean Composite Student Label

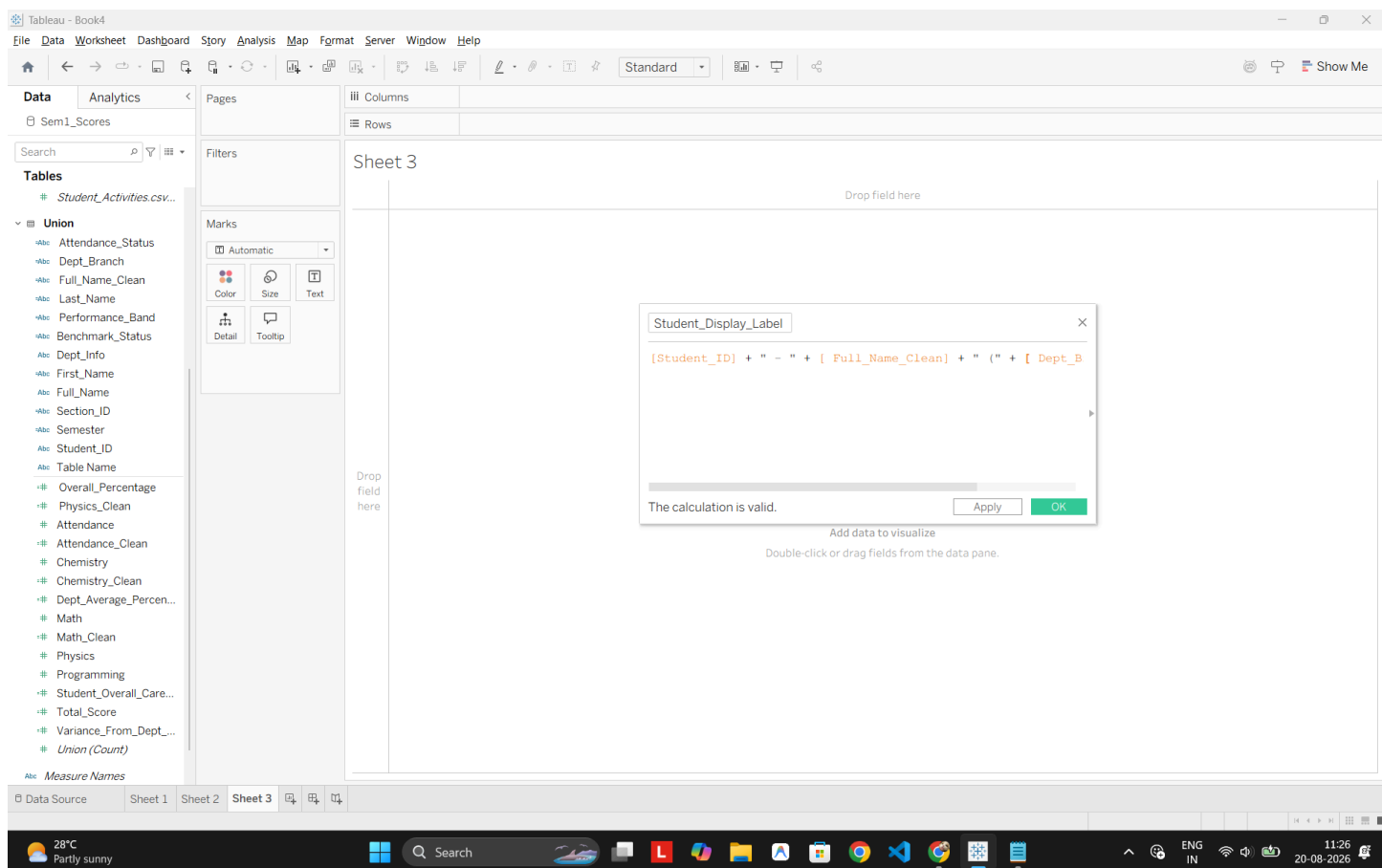
Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Field Name: Student_Display_Label
3. Tableau Formula:

Code snippet

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

4. Click OK.



Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

